

Single-photon avalanche diode imaging sensor for subsurface fluorescence LiDAR

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We realized subsurface fluorescence LiDAR with a large format single-photon avalanche diode array capable of localizing and quantifying concentration of fluorescent molecules in heavily scattering media, such as tissue, with submillimeter depth accuracy. © 2021 Optical Society of America under the terms of the

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Aside from surface topology mapping applications, large format single-photon avalanche diode (SPAD) imagers [1] are invaluable for macroscopic real-time 3D imaging and fluorescence lifetime microscopy (FLIM), owing to their ability to resolve the timing of single photons with picosecond accuracy, and with virtually zero readout noise. Challenges arise when the scene is composed of optically diffuse materials. If the optical photon transport is governed by strong multiple scattering events, then the time-of-flight (ToF) histograms required for volumetric reconstruction are convolved with diffuse reflectance dynamics that severely limit the spatial accuracy. Additionally, the distribution of optical photon mean free paths further accentuates the absorption effects that are problematic for quantitative imaging of signal sources within optically diffuse media. Unlike in diffuse optical tomography [2] (DOT), fluorescence depth sensing could be easier to perform by separating the interdependence of optical properties of medium and fluorophore distribution. The diffuse reflectance is indicative of the optical properties of the medium, and the fluorescence signal is indicative of both the fluorescent object's topology (depth), as well as the optical properties. Thus, the measurement of both the ToF information together with diffuse reflectance parameters should allow recovery of the location, shape, and concentration of fluorophore in the object. Recent rapid developments in picosecond-time-resolved large-format SPAD imagers [3] provide the ability to test this concept in a single-sensor, epi-illumination arrangement that could match the geometry used in medical imaging, microscopy, and metrology.

Our presented fluorescence LiDAR is a time-resolved optical imaging system that decodes the subsurface depth and concentration of fluorescent molecules using a single 512×512 SwissSPAD2 ToF SPAD sensor [3], in combination of either one or two light sources in an epi-illumination configuration (Fig. 1).

The same sensor records both fluorescence and diffuse reflectance images. We assume, simplistically, that the recorded temporal fluorescence signal I_f is a convolution of three components in time domain: (1) an instrument response function (IRF); (2) an impulse-response function of fluorophore f_{fl} (i.e., lifetime kinetics); and (3) an impulse-response function of radiation transport of excitation and emission light f_{tr} .

Assuming that IRF and f_{fl} are known and stable and can be readily deconvolved, we are left with the remaining diffuse transport response, which depends on four key parameters: absorption and reduced scattering coefficients (μ_a and μ'_s , respectively), volume distribution of the fluorophore, and time. Therefore, to solve the inverse problem for fluorophore distribution, μ_a and μ'_s should be evaluated independently. We extract these parameters from diffuse reflectance, while testing two different arrangements in the temporal and spatial domains, as shown in Fig. 1. In the first case, we record the temporally resolved diffuse reflectance using the same excitation laser while replacing the bandpass filter with a polarizing filter, and extract μ_{eff} from the map of the temporal response. In the second case, we keep the bandpass filter and use a secondary, structured light source, and demodulate the structured light pattern in the spatial domain to obtain μ_a and μ'_s . The latter spatial frequency-domain imaging (SFDI) [4] approach proves more robust, since switching the filter becomes unnecessary; it is also more accurate, at a cost of decreased spatial resolution of the μ_a and μ'_s map.

We realized this setup with a 635 nm picosecond wide-field illuminator (LDH 635, Picoquant, Germany) that worked in tandem

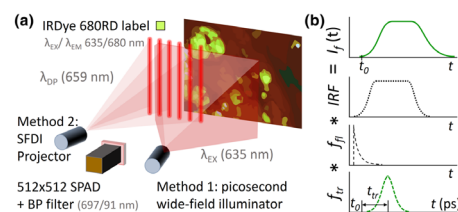


Fig. 1. (a) Experimental setup of fluorescence LiDAR for subsurface depth profiling. Temporal profiles in (b) illustrate components of detected signal $I_f(t)$: instrument response function IRF, lifetime kinetics f_{fl} , and photon propagation function f_{tr} with mean photon diffusion time t_{tr} and surface reflection time t_0 .

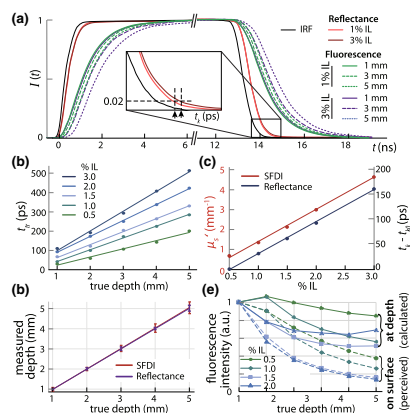


Fig. 2. (a) Temporal profiles of reflectance and fluorescence of IRDye 680RD inclusion at varying depth and intralipid (IL) concentration; (b) relation between rising edge delay and inclusion depth; (c) SFDI and reflectance-based measurements of μ_s' (at 659 nm) and t_k for varying IL concentration; (d) estimated depths of inclusion using both methods; (e) depth-corrected inclusion intensity.

with the SPAD sensor at 20 MHz pulse repetition rate. Near-infrared fluorophores (IRDye 680RD, $f_H = \exp[-t/0.7 \text{ ns}]$) were embedded in a scattering medium (Intralipid, IL) at different depths and varying μ_s' . The SPAD sensor recorded fluorescence image stacks at 17.8 ps temporal gate step over 25 ns temporal range. The maximum normalized intensity slew rate of the whole setup was $1/500 \text{ ps}^{-1}$, which yields ToF resolution of 0.8 mm assuming a moderate signal-to-noise ratio of 25. However, as demonstrated in Fig. 2, this resolution improved substantially for fluorescence signals embedded in scattering media, where the medium modulated the temporal fluorescence dynamics and therefore enhanced the depth sensitivity. We modeled the diffuse photon transport with a simple Gaussian model, following the notation in Ref. [4], yielding a mean time of t_{tr} . Next, we recorded a temporal stack of diffuse reflectance images, followed by signal normalization and mapping the times of t_{tr} and at 2% intensity threshold, which yielded both surface topology (t_{tr} map) and a diffuse reflectance kernel half-width (t_k map). In the second method, we extracted μ_a and μ_s' with an SFDI system (ReflectRS, Modulim, CA) using a frequency demodulation in the range of $0 - 0.2 \text{ mm}^{-1}$. In either case, we constructed a calibration lookup table using embedded fluorescent phantoms, where depth and μ_s' were controlled over physiological ranges between 0–5 mm and $0.1 - 4 \text{ mm}^{-1}$, respectively. Figure 2 shows the results of a 10 μM , 4 mL IRDye 680RD inclusion located at different depths and optical properties of the embedding medium.

In Fig. 2(a), we plot normalized temporal profiles of fluorescence for 1–5 mm inclusion depths and different phantom scattering coefficients. Figures 2(b) and 2(c) show extracted parameters—temporal delay t_0 and μ_s' for fluorescence and diffuse reflectance. Finally, depth calibration in Fig. 2(d) demonstrates the accuracy of 0.3 mm at 5 mm depth, as well as a linear relationship between calculated and actual depth of phantom inclusions. Figure 2(e) demonstrates the correction of the detected fluorescence intensity (“on surface”) to a predicted true intensity (“at depth”) for varying scattering tissue properties and depths.

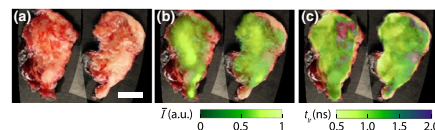


Fig. 3. Aligned composite of H&N tumor specimen photographs (a) overlaid with (b) standard (steady-state) fluorescence signal I and (c) map of fluorescence propagation time t_{tr} . Calibration bar 10 mm.

Finally, we show *in vivo* fluorescence LiDAR imaging of a resected head & neck (H&N) tumor tissue in Fig. 3, carried out in full compliance with local IRB protocol. The tumor was selectively labeled by intravenous injection of 180 nanomoles of solubilized ABY-029 (anti-epithelial growth factor receptor Affibody molecule labeled with IRDye 800CW, $f_H = \exp[-t/0.63 \text{ ns}]$ for bound fluorophore [5]). Figure 3(a) shows a cumulative fluorescence signal (interframe temporal SNR ≈ 15), while Fig. 3(b) shows the extracted diffuse ToF map of t_{tr} . The ToF map intensity closely follows the tissue blood concentration and type, which are expected to have varying μ_{eff} . Our purpose was to demonstrate the feasibility of ToF imaging detecting picomolar fluorophore concentration in a clinical microdose case, and the results warrant further controlled studies of labeled tumor margins.

This work is motivated by the pervasive desire of surgeons to see molecular information about tissue below the surface and distinguish malignancies from healthy tissue. It is possible to label tumors with specific fluorescent molecules, but not to easily see into tissue beyond microscopy limits of near a millimeter. This negatively affects the utility and objectivity of fluorescence guidance. Similar problems arise in through-fog imaging [4], wide-field FLIM [6], and other Mie scattering dominated scenarios. Our aim was to test large-format SPAD sensors in biomedical imaging and demonstrate their unique ability to image below the surface with diffuse radiation transport.

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